Ghost Data Still Exists: Deletion-Audit Agent

HackerRank Orchestrate, October 2026. Submission brief, 25 Oct 2026.

Approach

This is an offline, evidence-first auditor. It treats every workflow status as a **claim** and every evidence readback as a **fact** with `observed_at` and `available_at` times. It decides each case using only facts available at its `audit_at`, then applies `deletion_rules.csv` through a small policy interpreter. Two independent checkers must pass before `output.csv` is written. The prediction run makes **0 model calls**. LLMs classify ambiguous business documents (two models must agree; the result is cached and replayed offline) and review results as an advisory step.

Architecture

```
dataset/ ──► [1] Ingest: SQL→SQLite, JSON, logs, PDF text, OCR ──► immutable timestamped Facts
                 + cached two-model LLM document classifications (bound to file sha256)
          ──► [2] Intent: deletion authorized?  no → NOT_APPLICABLE, []
          ──► [3] Identity: current + historical email/phone/hash → user_id
          ──► [4] Resource closure: subject links, cross-refs, workflow accounts,
                  catalog links inside the case's own request window
          ──► [5] Cutoff filter: observed_at and available_at ≤ audit_at
          ──► [6] Policy interpreter: DELETE / REDACT / ANONYMIZE / RETAIN / BLOCK / DISABLE
          ──► [7] Planner: FINDING → REMEDIATION (flows, restores first) → VERIFICATION
          ──► [8] Independent checks: plan simulator (fixed point, no wrongful delete)
                  + validator (schema, order, citations ≤ audit_at, raw-file hash recheck)
          ──► output.csv  (written only when every row is supported)
```

Results

32 labeled samples	Value
Status accuracy	31/32
Exact full plans	20/32
Step precision / recall / F1	0.883 / 0.819 / 0.849
Field accuracy (evidence_id)	179/193
Ordering (canonical)	32/32
False COMPLETE · wrongful deletion	0 · 0

Full 518 run and QA	Value
Rows / unresolved	518 / 0
COMPLETE · INCOMPLETE · N/A	153 · 355 · 10
Invariant violations	0
Edge-case fixtures	28/30
Unit and regression tests	225 passed
Raw-evidence spot check (25)	20 ok · 1 fixed · 4 unsure

What moved the score

- **Unresolved does not mean missing data.** Over-strict assumptions blocked 442 → 207 → 5 → 0 cases. Each fix was justified by the labeled samples and recent sources (EDPB Feb 2026, NIST SP 800-88r2, ICO/CNIL).
- **Security logs are retained** without a per-record expiry; only an explicit expired date blocks.
- **Already-compliant records are verified,** for example accounts proven absent by a complete table inventory. Exact plans went from 9 to 20.
- **An unblocked backup is a restore risk; a found legal hold is retained;** catalog links count only inside the case's own request window.

How it was built: agent team with memory

Agent	Input	Output
researcher	question + contract	dated claims that ≥2 recent sources agree on
implementer	gap list + sample diffs	engine patch with before/after metrics; accepted only on a strict gain
qa (Harness 2.0)	code, samples, 30 fixtures, 518 run	scorecard, diffs vs the previous run, raw-evidence verification
reviewer	pipeline vs contract	ranked gaps, list of safety behaviours to keep
output	final run	code.zip, reports, clean-extract reproduction

Each agent reads a shared memory file and its own, and writes its result to `runs/agents/<name>/output.md`. Definitions and memories ship in `code.zip/agents/`.

Interview: likely questions

- **Why not trust "COMPLETE"?** Statuses are claims. Only readbacks count, and an absence needs a complete inventory.
- **Time travel?** Both timestamps must be ≤ audit_at, so later evidence can't change an earlier case.
- **Retention vs deletion?** The rules CSV decides. The simulator asserts no retained record is deleted (0 wrongful deletions).
- **Where do LLMs help?** Document classification: gpt-5.5 and gpt-5.4-mini must agree, and the result is cached against the file hash. Review is advisory only, and an LLM reviewer's "major" flags were refuted against raw files.
- **Weaknesses?** 118 rows lack an account check because no account id is linked; analytics records without identifiers are skipped rather than guessed. These cost steps, not statuses.
- **Cost?** 0 calls at prediction time; about 30k development/QA tokens, under $0.10.

Submission

Field	File
code.zip	`upload/code.zip`: code, prompts, tests, README, evaluation/, agents/, .env.example (no keys, no dataset)
output.csv	`upload/output.csv`: 518 rows
chat_transcript	`upload/log.txt` (refresh after closing the session: `runs/FINALIZE.sh`)
AI interview	this brief + `upload/INTERVIEW.md`

Submit at: https://www.hackerrank.com/contests/hackerrank-orchestrate-october26/challenges/ghost-data-still-exists